

Study on Classification of Left-Right Hands Motor Imagery EEG Signals Based on CNN

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Abstract—The performance of classification is one of the most key issues in brain computer interface (BCI) system. This paper proposes a classification method of two-class motor imagery electroencephalogram (EEG) signals based on convolutional neural networks (CNN), in which EEG signals from C3, C4 and Cz electrodes of publicly available BCI competition IV dataset 2b were used to test the performance of CNN. We investigate CNN with a form of input from short time Fourier transform (STFT) combining time, frequency and location information. Fisher discriminant analysis-type F-score based on band pass (BP) feature and power spectra density (PSD) feature are employed respectively to select the subject-optimal frequency bands. In the experiments, typical frequency bands related to motor imagery EEG signals, subject-optimal frequency bands and Extension Frequency Bands are employed respectively as the frequency range of the input image of CNN. The better classification performance of Extension Frequency Bands show that CNN can extract optimal feature from frequency information automatically. The classification result also demonstrates that the proposed approach is more competitive in prediction of left/right hand motor imagery task compared with other state-of-art approaches.

Keywords—*electroencephalogram; brain computer interface; motor imagery; convolutional neural networks; optimal frequency band; Fisher discriminant analysis-type F-score*

I. INTRODUCTION

A brain-computer interface (BCI) is a hardware and software system that can communicate with computer or external devices through brain activity [1]. BCI system can provide a new human computer interaction method which does not depend on the traditional peripheral nervous system and muscle tissue such as a mental typewriter presented by Berlin BCI [2] and an approach based on recognition of sign language from imagination [3]. BCI is also a promising technology for the communication of locked-in state (CLIS) patients [1, 4] as well as the interaction with videogames or virtual environments [5].

A key phenomenon observed across different recording modalities is that the neurophysiological rhythmic activities, mostly in alpha (7-13 Hz), mu (8-12 Hz), beta (14-30 Hz) and gamma

frequency bands (>30 Hz), recorded over the sensorimotor cortex are modulated by actual movement, motor intention, or motor imagery. Such rhythmic brain activities measured by EEG or other recording modalities over the sensorimotor cortex are collectively referred to as the sensorimotor rhythms (SMRs) [6]. Motor imagery (MI) EEG signals is a kind of original EEG signals containing SMRs. MI-EEG signals one of the most popular EEG signals used in non-invasive BCI studies to serve as control signals. Different MI tasks can cause different oscillatory activities observed in the sensorimotor cortex of the brain in the form of event-related desynchronization (ERD) or event-related synchronization (ERS) [7].

Nevertheless, compared with such traditional interaction modes as keyboard and mouse, the precision and reliability of BCI system are still insufficient since the performance varies greatly across or even within the subjects [8]. EEG signals are often distorted by such artifacts as electromyography (EMG) or electrooculography (EOG) [1]. Moreover, in most of the cases, EEG performance is a typical non-stationary process. Therefore, it is still a significant challenge to decode mental activities from EEG signals accurately.

In recent years, with the increase of the size of data, deep learning algorithm is known to provide better classification performance in such fields as shape recognition and speech recognition. In the study on BCI, An et.al [9] construct a deep belief net (DBN) for MI EEG signals classification and their results is better than those from SVM. Yang et.al [10] also employ convolutional neural networks (CNN) to classify the augmented common spatial pattern (ACSP) features of MI-EEG signals. In recent study, Tabar et.al [11] investigate convolutional neural networks and stacked autoencoders (SAE) which have better classification performance compared with state-of-art approaches for left/right hand MI-EEG signals classification.

Inspired by the success of CNN-SAE proposed by Tabar et.al [11], in order to reduce the parameters of neural network and improve the performance, we extend their work of CNN on the classification of motor imagery EEG signals. The proposed approach is analyzed and evaluated by using BCI Competition IV dataset 2b [12].

In this paper, the processing method of input data of CNN combining time, frequency and location information is introduced. First, the original signals obtained from C3, Cz and C4 electrodes are preprocessed to extract useful information of MI-EEG signals. Then Short Time Fourier Transform (STFT) is employed to convert the preprocessed data from each electrode to 2-dimensional feature matrix. The data of feature matrix of related frequency bands (i.e. mu band and beta band) from each electrode is extracted and combined to one matrix which consists of the input data of CNN.

In our study, 1-dimensional convolutional neural network is constructed and trained by the input images to learn the activation pattern. Since CNN only contains one convolutional layer and one max-pooling layer, this significantly reduces the parameters of network compared with CNN-SAE and the optimal neural network can be easily developed to improve the classification performance.

To improve the performance of CNN, typical mu band and beta band related to MI task, optimal frequency bands selected by fisher discriminant analysis-type F-score based on band pass (BP) feature and power spectra density (PSD) feature, and Extension Frequency Bands are employed on BCI Competition IV dataset 2b to select the optimal frequency band of input data of CNN.

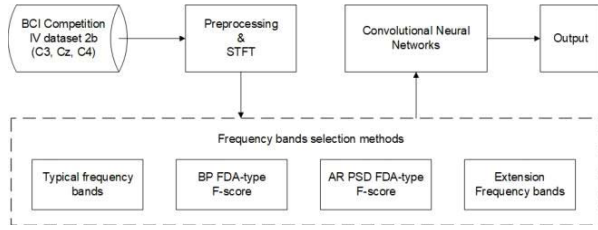


Fig. 1. The proposed approach for two-class MI classification based on CNN

II. METHOD

Source imaging studies of SMRs have revealed that movement or motor imagery of different body parts were associated with decrease in SMR from regions along the primary sensorimotor cortex corresponding to different body parts, known as the Homunculus [6]. Usually, the cerebral cortex near electrode C3 and electrode C4 is considered to be highly related to right hand movement and left hand movement respectively. Cerebral cortex near electrode Cz will also be affected by hands movement. Our work is focus on the EEG signals detected from these three electrodes to predict left/right hand movement.

Figure 1 illustrates the diagram of the proposed approach for two-class MI classification based on CNN, in which STFT is first employed to the preprocessed data from C3, Cz and C4 electrodes to construct the feature matrix. Then the feature matrix was reshaped to the input image of CNN by different frequency band selection methods. After obtaining the input images, 1-dimensional convolutional neural network was trained and employed to classify MI-EEG signals.

A. Preprocessing

Public left/right hand MI-EEG task dataset (BCI Competition IV dataset 2b [12]) is chosen in our study to analyze the

performance of the proposed method. A practical issue in EEG dataset is that the size of data usually is not large enough to learn thousands or even millions of parameters in deep learning based approaches. In our study, a four-second trial (from 3 s to 7 s) related to motor imagery task from dataset is divided into eleven segments of two-second length with an overlap of 90% to increase the input data size. Each segment from three electrodes is preprocessed using the following averaging method:

$$y_i^{Cx} = x_i^{Cx} - \frac{1}{N} \sum_{i=1}^N x_i^{Cx} \quad (1)$$

in which N is the size of samples from each two-second segment, x_i^{Cx} denotes the i th sample of each segment from electrode Cx (C3, C4 or Cz), and y_i^{Cx} is the result of preprocessing.

The aim of this process is to eliminate the difference average intensity of EEG signals from different electrode and make them contribute to the input image of CNN equally. After preprocessing, the preprocessed segments only contain oscillatory information of EEG signals.

B. Feature Matrix

Typically the energy in mu band (8–13 Hz) collected in sensorimotor cortex of the brain will decrease when performing an MI task. This decrease is termed as Event Related Desynchronization (ERD) [13]. An energy increase termed as Event Related Synchronization (ERS) in the beta band (13–30 Hz) occurs after an MI task. Left/right hand MI tasks can cause ERD and ERS respectively in the right and left sides of the sensorimotor cortex, which will affect EEG signals at C4 and C3 electrodes [11, 14]. Also, EEG signals at Cz are affected by hands movement MI task.

Considering these SMRs characteristics of MI-EEG signals, STFT is chosen to process each preprocessed two-second length segment obtained from each electrode. The segment that contains 500 samples on account of the sample frequency of the dataset is 250 Hz. STFT is performed with window size of 64 and time lapses of 14, leading to a feature matrix with the size of $N_F \times 32$, in which 32 is the length of data along the time axes, N_F is the length of data along the frequency axes and N_F is determined by different frequency band selection methods which will be discussed in the following part. The number of fast Fourier transform (FFT) point of STFT is chosen to be 512.

C. Frequency band selection method

1) Typical mu band and beta band

Mu band (8-13 Hz) [7] and beta band (13-30 Hz) [15] are usually considered to be related with MI task. In the study of Tabar et.al [11], the beta band between 17-30 Hz is considered to be more related to left/right hand MI task. By selecting typical frequency band (8-13 Hz and 17-30 Hz), the size of feature matrix of mu band is 12×32 and the size of feature matrix of beta band is 29×32 . To make the two bands equally contribute to the input images of CNN, beta band is resized to 12×32 by using cubic interpolation method. Finally, these matrices of each band from different electrodes are combined to a $N_{fr} \times 32$ feature matrix where $N_{fr} = N_c \times 2 \times 12 = 72$. $N_c = 3$, which represents the number of electrodes. N_{fr} is the vertical size (frequency domain) of the input image of CNN (Figure 2).

2) BP FDA-type F-score

One of the most difficult improvements of classification is the difference among subjects [16]. Different subjects always have different specific frequency bands related to MI task. In order to solve this problem, Yang et.al [17, 18, 19] proposed a method of Fisher discriminant analysis-type F-score based on band pass (BP) feature to simultaneously select the optimal frequency band and optimal time segments of each subject to improve the classification performance of MI task.

In our study, BP feature is defined as:

$$BP = \text{mean}(\log(\text{var}(i))) \quad (2)$$

where $\text{var}(i)$ denotes the variance for two-second length segment i , $\text{mean}(\cdot)$ denotes the mean value of data. Logarithm is also applied to make the variance data distributions approximately normal.

Since the feature vector, $[BP_{C3}, BP_{Cz}, BP_{C4}]$ contains the BP features from three electrodes C3, Cz and C4, the one-versus-rest (OVR) based F-score is calculated by:

$$\hat{F} = \frac{(\overline{BP}_{C3}^o - \overline{BP}_{C3}^{c(i)})^2 + (\overline{BP}_{Cz}^o - \overline{BP}_{Cz}^{c(i)})^2 + (\overline{BP}_{C4}^o - \overline{BP}_{C4}^{c(i)})^2}{(\overline{S}_{C3}^o + \overline{S}_{Cz}^o + \overline{S}_{C4}^o) + (\overline{S}_{C3}^{c(i)} + \overline{S}_{Cz}^{c(i)} + \overline{S}_{C4}^{c(i)})} \quad (3)$$

in which

$$\overline{BP} = \frac{1}{K} \sum_{k=1}^K BP(k) \quad (4)$$

$$\bar{S} = \frac{1}{K-1} \sum_{k=1}^K (BP(k) - \overline{BP})^2 \quad (5)$$

where K is the number of segments.

In our study, BP FDA-type F-score subject-optimal frequency bands are selected from 4-14 Hz and 16-40 Hz respectively, which represent the range of mu band and beta band. In the range of 4-14 Hz, the sliding windows with a width of 4, 5, 6 Hz and an interval of 1 Hz is employed successively to find the subject-optimal mu band corresponding to the max F-score. In the range of 16-40 Hz, the sliding windows with a width of 8, 10, 12 Hz and an interval of 1 Hz is employed successively to find the subject-optimal beta band. Sixth order Butterworth filter is applied to the segments to get the filtered data. The result of subject-optimal frequency band of each subject in BCI Competition IV dataset 2b is presented in Table I.

3) AR PSD FDA-type F-score

Although BP features can be extracted quickly, the feature matrix focuses on power information in our study. Auto-regressive model (AR) power spectrum analysis has been widely used in power feature extraction from MI-EEG signals [20, 21]. According to the study of Lin et.al [20], 12th order AR model has better performance in MI-EEG signals classification compared with other orders.

In our study, 12th order AR model based on Burg algorithm [22] is employed to extract the AR PSD feature by estimating the power spectra density (PSD) of each two-second length segment. The number of fast Fourier transform (FFT) point of AR model PSD estimation based on Burg algorithm is 1000. Different from BP FDA-type F-score, AR PSD feature, extracted by equation (6), is used to replace BP feature:

$$F_{PSD} = \text{mean}(\log(\text{var}(PSD_i))) \quad (6)$$

where PSD_i denotes AR PSD feature extracted from the value of power density of mu band or beta band. The selection range of frequency bands and the set of sliding window is the same as in part 2) of this section. The only difference is that we use F_{PSD} instead of BP . Table I presents the result of subject-optimal frequency bands based on AR PSD FDA-type F-score for left hand and right hand MI task of BCI competition IV dataset 2b.

4) Extension frequency band

The average accuracy of subject-optimal frequency bands by using the method of BP FDA-type F-score and AR PSD FDA-type F-score is not improved compared with the average accuracy resulted from typical frequency bands related to MI task. In the result of subject-optimal frequency bands of BP FDA-type F-score, the mu band of subjects and beta band of subjects are 4-14 Hz and 17-36 Hz respectively. In the result of subject-optimal frequency bands of AR PSD FDA-type F-score, the range of optimal mu band and the range of optimal beta band of all subjects are 6-14 Hz and 16-38 Hz respectively. Refer to the result of classification in section 3, we also propose Extension Frequency Band in which the range of mu band is 4-14 Hz and the range of beta band is 16-32 Hz to extract more feature information related to MI task. In this way, $N_{fr} = 132$ and size of feature matrix is $N_{fr} \times 32$ by using cubic interpolation method on beta band.

TABLE I. OPTIMAL FREQUENCY BAND USING BP FDA-TYPE F-SCORE AND AR PSD FDA-TYPE F-SCORE

Subject	Optimal Frequency Band (Hz)			
	BP FDA-TYPE F-SCORE		AR PSD FDA-TYPE F-SCORE	
	mu band	beta band	mu band	beta band
1	10-14	17-25	8-14	26-38
2	6-11	17-27	8-14	23-35
3	4-8	28-36	10-14	25-33
4	10-14	19-27	10-14	16-24
5	10-14	23-31	6-10	26-38
6	10-14	20-28	10-14	16-28
7	10-14	20-28	10-14	16-24
8	9-14	29-37	10-14	16-24
9	10-14	20-28	7-13	24-36

BCI Competition IV dataset 2b

D. CNN

Convolutional neural networks (CNN) [23] is a class of deep, feed-forward artificial neural networks that has successfully been applied to recognize shapes in images and CNN is partially invariant to the location of shapes. A CNN consists of an input layer, multiple hidden layers and an output layer. The hidden layer of CNN typically has convolutional layers and pooling layers. The parameters of network and the filters of convolutional layers are learned through back-propagation algorithm [24] in order to improve the performance of classification.

In order to improve the convergence performance of cost function of CNN and reduce the effect of artifact, the element value of input images of CNN can be calculated as:

$$I_i = \begin{cases} 1 & M_i > 0.95 \times \max(M_i) \\ \frac{M_i}{0.95 \times \max(M_i)} & M_i \leq 0.95 \times \max(M_i) \end{cases} \quad (7)$$

in which M_i is the element value of feature matrix, for $i = 1, 2, \dots, N_{fr} \times 32$; $\max(\cdot)$ denotes the max element value of feature matrix. The value of N_{fr} depend on the frequency bands selection method.

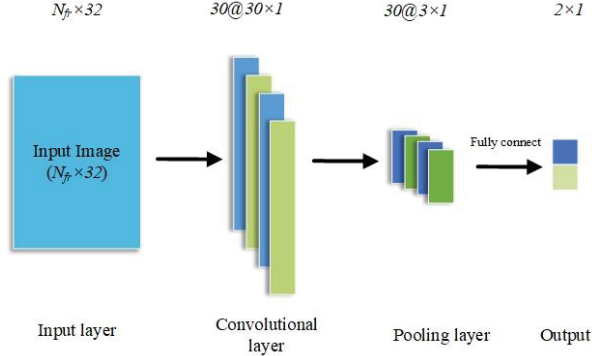


Fig. 2. Architecture of proposed CNN (the column size N_{fr} of input image is determined by different frequency band selection method)

The proposed CNN structure is shown in Figure 2. We use one convolutional layer and one max-pooling layer to construct the hidden layers of CNN. The input image of CNN in our study contains time, frequency and location information. Frequency and location information in vertical location plays an important role in the classification performance. $N_f = 30$ 1-dimensional filters with the size of $N_h \times 3$ are firstly applied to the input image along time axis in convolutional layer, in which N_h is equal to N_{fr} and depends on the feature selection methods. The convolution step is one and N_h denotes the vertical size of the input image. Then the result of convolution is fed to the activation function to generate the output feature maps. The k th feature map at a given layer can be obtained as:

$$h_{ij}^k = f(a) = f((W^k * x) + b_k) \quad (8)$$

where x is the input image, W^k is the weight matrix for filter k and b_k is the bias value, for $i = 1, 2, \dots, N_{fr}$. The activation function f is rectified linear unit (ReLU) function. ReLU is approximated by the following softplus function:

$$f(a) = \text{ReLU}(a) = \ln(1 + e^a) \quad (9)$$

where a is defined in Equation (8).

The convolutional layer in the proposed approach is followed by a max-pooling layer and a single fully-connected layer. Ten sampling factor and zero padding is applied in max-pooling layer. The output maps of max-pooling layers are subsampled into N_f vectors with the size of 3×1 . The fully-connected layer has two outputs corresponding to left hand and right hand MI task. Parameters of the CNN are learned by using back-propagation algorithm. Gradient descent method is applied to minimize the following error:

$$C = 0.5 * (y - a)^2 \quad (10)$$

where y is the ideal output and a is the predicted output of CNN. Finally, the trained CNN is employed in the classification of the two-second length segments in test set.

TABLE II. ACCURACY (%) RESULTS OF CNN BASED ON DIFFERENT FREQUENCY BANDS SELECTION METHOD

Subject	Accuracy (%)			
	Typical Frequency band	BP FDA-type F-score	AR PSD FDA-type F-score	Extension Frequency Band
1	88.04	84.42	88.04	89.57
2	79.14	81.66	73.22	86.83
3	80.45	80.16	73.48	82.73
4	95.37	93.06	93.20	95.51
5	89.63	90.32	90.03	93.22
6	84.77	85.85	79.08	83.69
7	87.76	80.97	86.50	90.58
8	87.62	87.89	87.89	91.16
9	86.79	87.90	87.76	88.32
Average	86.62	85.50	84.36	89.07

BCI Competition IV dataset 2b

III. EXPERIMENTAL RESULT AND DISSICUSSION

A. Dataset

Public left/right hand MI-EEG BCI Competition IV dataset 2b [12] is analyzed and evaluated in the experiment. Five sessions were recorded from each subject. Each session comprised EEG data recorded from 3 bipolar recordings (C3, Cz, and C4) and 3 monopolar EOG channels. The training data, which consisted the first 2 sessions and the 3rd session, comprised 240 trials without visual feedback and 160 trials with visual feedback respectively. The evaluation data consisted 2 sessions of EEG data that comprised a total of 320 trials.

B. Result and Discussion

In each trial of dataset, totally eleven segments of two-second length with an overlap of 90% in MI part of each trial (3-7s) are obtained from each electrode (C3, C4 or Cz) are extracted and preprocessed. All the trial of five sessions are processed and evaluated equally in our study.

After preprocessing and the process of STFT, the input images containing time, frequency and location information are constructed by different frequency band selection methods. Then CNN is trained and tested separately for each subject as the classifiers. The architecture of CNN is shown in Figure 2. In our study, the size of filter in the convolutional layer is $N_{fr} \times 3$ and N_{fr} is determined by the proposed feature frequency bands selection methods. The network is trained by using batch training method with batch size equal to 40 for 1400 epochs. The performance of the proposed approach is evaluated by the classification accuracy using 10-fold cross-validation method. In this way, for each subject, 90% of the input images are selected randomly as the training set and the remaining 10% are selected as the test set. The result of classification of different frequency bands selection method is shown in Table II.

Compared with the typical frequency band selection method, the average performance of classification of subject-optimal frequency band selection method (BP FDA-type F-score and AR FDA-type F-score) are not improved. However, the classification accuracy of subject 2, 5, 6, 9 are improved by using BP FDA-type F-score and the classification accuracy of subject 5, 9 are also improved by using AR PSD FDA-type F-score. Since the optimal mu band of subject 4 and beta band of subject 5 are not contained in the range of typical frequency bands, we propose to extend frequency bands 4-14 Hz and 16-32 Hz to extract more frequency features of subjects in BCI Competition IV dataset 2b to improve the performance. It should be noted that larger range will increase the learning parameters of filters, which makes the parameters of CNN more difficult to learn. The result of CNN based on Extension Frequency Bands is also shown in Table II. The notable performance improvement of Extension Frequency Band selection methods indicates that CNN can potentially select subject-optimal frequency feature with wider range of input frequency band.

The result of CNN based on Extension Frequency Bands in our study is also compared with those from the current state of art studies and the kappa value is presented in Table III. Kappa value is calculated to represent the classification accuracy by removing the effect of random classification:

$$\text{kappa} = \frac{p_0 - p_e}{1 - p_e} \quad (11)$$

where p_0 is the overall agreement, which is the same as the classification accuracy and the chance agreement denote by p_e representing the case when the classification is purely by chance.

Compared with FBCSP [25] (winner methods of BCI Competition IV), Twin SVM [26] and CNN-SAE [11], the proposed CNN approach based on Extension Frequency Bands (EFB) has an obvious improvement of more than 40%. This can be partially contributed to the different sampling selection methods. In the approaches based on FBCSP, Twin SVM and CNN-SAE, in each trial, only one sample is extracted as train sample or test sample. Whereas in our study, totally 11 samples are extracted with 90% overlap for each trial and all samples are combined and separated randomly to the training set and test set. In this way, part of the samples for testing is similar to some samples for training. This will also result in that some input images of CNN in the training set are similar to the input images in test set, which can lower the difficulty of classification.

However, in our study the size of input images in the training set is about 6500 for each subject. For each input image in the test set, at most 9 input images in the training set is similar, which is less than 0.14% of the size of input images in the training set, and the same part of the original samples are from 10% to 90%. Thus, the similar part of samples only has a slight effect for the training of CNN. On the other hand, the good performance of the proposed approach also indicates that the proposed method is suitable for training dataset with a large number of different data and a small amount of the similar data.

To compare with other approaches for fairness, we also perform trial-to-trial classification on BCI Competition dataset 2b. 400 trials from first three sessions in dataset are separated randomly to training set and test set by using 10-fold cross-validation method. Each trial in test set is firstly classified as the

class to which most of the two-second length segments within that trial is predicted. For example, if more than 5 segments out of 11 segments in one trial are predicted to left-hand MI task, the trial will be labeled as left-hand MI task. Otherwise, it will be labeled as right-hand MI-task. Next, the trial-to-trial kappa value of each subject is calculated according to the classification result of trials of each subject. The kappa value of trial-to-trial by CNN based on Extension Frequency Bands is shown in Table III.

TABLE III. KAPPA VALUE RESULTS OF CNN BASED ON EXTENSION FREQUENCY BANDS METHODS COMPARED WITH FBCSP [25], TWIN SVM [26], CNN-SAE [11]

Subject	KAPPA Value				
	<i>EFB-CNN</i>	<i>FBCSP</i>	<i>Twin SVM</i>	<i>CNN-SAE</i>	<i>EFB-CNN (trial to trial)</i>
1	0.791	0.546	0.494	0.517	0.495
2	0.737	0.208	0.416	0.324	0.121
3	0.655	0.244	0.322	0.494	0.195
4	0.910	0.888	0.897	0.905	0.944
5	0.865	0.692	0.722	0.655	0.747
6	0.674	0.534	0.405	0.579	0.594
7	0.812	0.409	0.466	0.488	0.600
8	0.823	0.413	0.477	0.494	0.588
9	0.766	0.583	0.503	0.463	0.496
Average	0.781	0.502	0.526	0.547	0.531

BCI Competition IV dataset 2b

In trial-to-trial performance, the network is trained by using batch size equal to 40 for 300 epochs, which perform better in our experiment. In the result of our proposed approach, subject 4, 5, 6, 7, 8 show better performance compared with other methods. However, the classification accuracy of subject 2, 3 perform badly. Subject 2 and subject 3 also have poor performance in other methods compared with most of other subjects, which demonstrates that the proposed approach still needs to be improved for the subject with no apparent feature in MI-EEG task. The average kappa value of all subjects of the proposed approach is better than that of FBCSP and Twin SVM, and is slightly worse than that CNN-SAE. In contrast, 5/9 of all subjects have better performance when applying the proposed method, which demonstrates the competitive predictive performance of the proposed approach in the classification of left-right hands MI-EEG task.

Furthermore, the proposed method only extracts EEG signals from C3, C4 and Cz electrodes, which enables it to be easily employed in MI-BCI application.

IV. CONCLUSION

In this paper, two optimal frequency bands selection methods, BP FDA-type F-score and AR PSD FDA-type F-score, are employed to improve the performance of CNN. The classification result of these two methods is compared with that of CNN based on typical frequency bands. By analyzing the classification accuracy of different subjects in the dataset, we propose Extension Frequency Bands (mu band 4-14 Hz, beta band 16-32 Hz) to extract more features for the subjects in BCI Competition IV. The result shows that CNN can potentially select subject-optimal frequency feature with enough frequency range

of input image in our study. Compared with other state of art method, the better performance of our proposed approach illustrates that CNN based on Extension Frequency Bands can provide impressive performance in left-right hands MI EEG signals classification with the increased size of training set and less similar data in the training set. What is more, without similar data in the training set, our proposed approach also shows competitive predictive performance, in which 5/9 subjects have better performance compared with other state-of-art approaches.

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